

Appendix

Additional Material

A.1 Properties of Gaussian Distribution

Definition A.1 (Gaussian distribution). *A random variable $\mathbf{x} \in \mathbb{R}^n$ has a Gaussian distribution with mean $\mathbf{m} \in \mathbb{R}^n$ and covariance $\mathbf{P} \in \mathbb{R}^{n \times n}$ if its probability density has the form*

$$N(\mathbf{x} | \mathbf{m}, \mathbf{P}) = \frac{1}{(2\pi)^{n/2} |\mathbf{P}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mathbf{m})^\top \mathbf{P}^{-1} (\mathbf{x} - \mathbf{m})\right), \quad (\text{A.1})$$

where $|\mathbf{P}|$ is the determinant of the matrix $\mathbf{P}$.

Lemma A.2 (Joint distribution of Gaussian variables). *If random variables $\mathbf{x} \in \mathbb{R}^n$ and $\mathbf{y} \in \mathbb{R}^m$ have the Gaussian probability distributions*

$$\begin{aligned} \mathbf{x} &\sim N(\mathbf{m}, \mathbf{P}), \\ \mathbf{y} | \mathbf{x} &\sim N(\mathbf{H}\mathbf{x} + \mathbf{u}, \mathbf{R}), \end{aligned} \quad (\text{A.2})$$

then the joint distribution of $\mathbf{x}, \mathbf{y}$ and the marginal distribution of $\mathbf{y}$ are given as

$$\begin{aligned} \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} &\sim N\left(\begin{pmatrix} \mathbf{m} \\ \mathbf{H}\mathbf{m} + \mathbf{u} \end{pmatrix}, \begin{pmatrix} \mathbf{P} & \mathbf{P}\mathbf{H}^\top \\ \mathbf{H}\mathbf{P} & \mathbf{H}\mathbf{P}\mathbf{H}^\top + \mathbf{R} \end{pmatrix}\right), \\ \mathbf{y} &\sim N(\mathbf{H}\mathbf{m} + \mathbf{u}, \mathbf{H}\mathbf{P}\mathbf{H}^\top + \mathbf{R}). \end{aligned} \quad (\text{A.3})$$

Lemma A.3 (Conditional distribution of Gaussian variables). *If the random variables $\mathbf{x}$ and $\mathbf{y}$ have the joint Gaussian probability distribution*

$$\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} \sim N\left(\begin{pmatrix} \mathbf{a} \\ \mathbf{b} \end{pmatrix}, \begin{pmatrix} \mathbf{A} & \mathbf{C} \\ \mathbf{C}^\top & \mathbf{B} \end{pmatrix}\right), \quad (\text{A.4})$$

then the marginal and conditional distributions of $\mathbf{x}$ and $\mathbf{y}$ are given as follows:

$$\begin{aligned}\mathbf{x} &\sim N(\mathbf{a}, \mathbf{A}), \\ \mathbf{y} &\sim N(\mathbf{b}, \mathbf{B}), \\ \mathbf{x} | \mathbf{y} &\sim N(\mathbf{a} + \mathbf{C} \mathbf{B}^{-1} (\mathbf{y} - \mathbf{b}), \mathbf{A} - \mathbf{C} \mathbf{B}^{-1} \mathbf{C}^T), \\ \mathbf{y} | \mathbf{x} &\sim N(\mathbf{b} + \mathbf{C}^T \mathbf{A}^{-1} (\mathbf{x} - \mathbf{a}), \mathbf{B} - \mathbf{C}^T \mathbf{A}^{-1} \mathbf{C}).\end{aligned}\tag{A.5}$$

A.2 Block Matrix Inverses and Matrix Inversion Formulas

Block matrix inversion formulas (see, e.g., Lütkepohl, 1996) and matrix inversion formulas (sometimes called Woodbury's identities) are useful in simplifying various expressions involving Gaussian distributions. The block matrix inversion formulas are the following, and they are easy to verify by direct calculation.

Theorem A.4 (Block matrix inverses). *When $\mathbf{A}$ and $\mathbf{D} - \mathbf{C} \mathbf{A}^{-1} \mathbf{B}$ are invertible, then*

$$\begin{aligned}& \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix}^{-1} \\ &= \begin{pmatrix} \mathbf{A}^{-1} + \mathbf{A}^{-1} \mathbf{B} (\mathbf{D} - \mathbf{C} \mathbf{A}^{-1} \mathbf{B})^{-1} \mathbf{C} \mathbf{A}^{-1} & -\mathbf{A}^{-1} \mathbf{B} (\mathbf{D} - \mathbf{C} \mathbf{A}^{-1} \mathbf{B})^{-1} \\ -(\mathbf{D} - \mathbf{C} \mathbf{A}^{-1} \mathbf{B})^{-1} \mathbf{C} \mathbf{A}^{-1} & (\mathbf{D} - \mathbf{C} \mathbf{A}^{-1} \mathbf{B})^{-1} \end{pmatrix}.\end{aligned}\tag{A.6}$$

When $\mathbf{D}$ and $\mathbf{A} - \mathbf{B} \mathbf{D}^{-1} \mathbf{C}$ are invertible, then

$$\begin{aligned}& \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix}^{-1} \\ &= \begin{pmatrix} (\mathbf{A} - \mathbf{B} \mathbf{D}^{-1} \mathbf{C})^{-1} & -(\mathbf{A} - \mathbf{B} \mathbf{D}^{-1} \mathbf{C})^{-1} \mathbf{B} \mathbf{D}^{-1} \\ -\mathbf{D}^{-1} \mathbf{C} (\mathbf{A} - \mathbf{B} \mathbf{D}^{-1} \mathbf{C})^{-1} & \mathbf{D}^{-1} + \mathbf{D}^{-1} \mathbf{C} (\mathbf{A} - \mathbf{B} \mathbf{D}^{-1} \mathbf{C})^{-1} \mathbf{B} \mathbf{D}^{-1} \end{pmatrix}.\end{aligned}\tag{A.7}$$

We can derive various useful formulas for matrices by matching the blocks in the representations in Equations (A.6) and (A.7). For example, the following formulas are sometimes useful in deriving Gaussian conditioning, and Kalman filtering and smoothing equations.

Corollary A.5 (Matrix inversion lemmas). *By putting $\mathbf{A} = \mathbf{P}^{-1}$, $\mathbf{D} = \mathbf{R}$, $\mathbf{B} = -\mathbf{H}^\top$, and $\mathbf{C} = \mathbf{H}$, and matching the top left blocks in Equations (A.6) and (A.7), we get*

$$(\mathbf{P}^{-1} + \mathbf{H}^\top \mathbf{R}^{-1} \mathbf{H})^{-1} = \mathbf{P} - \mathbf{P} \mathbf{H}^\top (\mathbf{R} + \mathbf{H} \mathbf{P} \mathbf{H}^\top)^{-1} \mathbf{H} \mathbf{P}, \quad (\text{A.8})$$

and from the top right block, we get

$$(\mathbf{P}^{-1} + \mathbf{H}^\top \mathbf{R}^{-1} \mathbf{H})^{-1} \mathbf{H}^\top \mathbf{R}^{-1} = \mathbf{P} \mathbf{H}^\top (\mathbf{R} + \mathbf{H} \mathbf{P} \mathbf{H}^\top)^{-1}. \quad (\text{A.9})$$

A.3 Cholesky Factorization and Its Derivative

The Cholesky factor of the symmetric positive definite matrix $\mathbf{P}$ is a lower triangular matrix $\mathbf{A}$ such that

$$\mathbf{P} = \mathbf{A} \mathbf{A}^\top. \quad (\text{A.10})$$

The matrix $\mathbf{A}$ can be computed by the Cholesky factorization algorithm (see, e.g., Golub and van Loan, 1996) presented below.

Algorithm A.6 (Cholesky factorization). *The Cholesky factor $\mathbf{A}$ of the matrix $\mathbf{P} \in \mathbb{R}^{n \times n}$ can be computed as follows:*

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procedure CHOL( $\mathbf{P}$ )
  for  $i \leftarrow 1 \dots n$  do
     $A_{ii} = \sqrt{P_{ii} - \sum_{k < i} A_{ik}^2}$ 
    for  $j \leftarrow i + 1 \dots n$  do
       $A_{ji} = (P_{ji} - \sum_{k < i} A_{jk} A_{ik}) / A_{ii}$ 
    end for
  end for
  return  $\mathbf{A}$ 
end procedure

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The partial derivative of the Cholesky factor $\partial \mathbf{A} / \partial \theta$ with respect to a scalar parameters θ can be computed using the following algorithm once $\mathbf{P}$ and $\partial \mathbf{P} / \partial \theta$ are known. The algorithm can be derived by formally differentiating the Cholesky factorization algorithm equations.

Algorithm A.7 (Partial derivative of Cholesky factorization I). *The partial derivative $\mathbf{D} = \partial \mathbf{A} / \partial \theta$ of the Cholesky factor of $\mathbf{P} \in \mathbb{R}^{n \times n}$ with respect to a scalar parameter θ can be computed as follows:*

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procedure DCHOL( $\mathbf{P}, \partial \mathbf{P} / \partial \theta$ )
  for  $i \leftarrow 1 \dots n$  do

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$$A_{ii} \leftarrow \sqrt{P_{ii} - \sum_{k < i} A_{ik}^2}$$


$$D_{ii} \leftarrow (\partial P_{ii} / \partial \theta - \sum_{k < i} 2 D_{ik} A_{ik}) / A_{ii} / 2$$

for  $j \leftarrow i + 1 \dots n$  do
   $A_{ji} \leftarrow (P_{ji} - \sum_{k < i} A_{jk} A_{ik}) / A_{ii}$ 
   $\text{temp} \leftarrow \partial P_{ji} / \partial \theta - \sum_{k < i} (D_{jk} A_{ik} + A_{jk} D_{ik})$ 
   $D_{ji} \leftarrow \text{temp} / A_{ii} - (D_{ii} / A_{ii}) A_{ji}$ 
end for
end for
return  $\mathbf{D}$ 
end procedure

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Another way to compute the same derivative is via the following theorem.

Theorem A.8 (Partial derivative of Cholesky factorization II). *The partial derivative $\partial \mathbf{A} / \partial \theta$ of the lower triangular Cholesky factor $\mathbf{A}$ such that $\mathbf{P} = \mathbf{A} \mathbf{A}^\top$ with respect to a scalar parameter θ can be computed as*

$$\frac{\partial \mathbf{A}}{\partial \theta} = \mathbf{A} \Phi \left(\mathbf{A}^{-1} \frac{\partial \mathbf{P}}{\partial \theta} \mathbf{A}^{-\top} \right), \quad (\text{A.11})$$

where $\Phi(\cdot)$ is a function returning the lower triangular part and half the diagonal of the argument as follows:

$$\Phi_{ij}(\mathbf{M}) = \begin{cases} M_{ij}, & \text{if } i > j, \\ \frac{1}{2} M_{ij}, & \text{if } i = j, \\ 0, & \text{if } i < j. \end{cases} \quad (\text{A.12})$$

Proof We use a similar trick that was used in the derivation of the time derivative of the Cholesky factor in Morf et al. (1978). We have

$$\mathbf{P} = \mathbf{A} \mathbf{A}^\top. \quad (\text{A.13})$$

By taking the derivative with respect to θ , we get

$$\frac{\partial \mathbf{P}}{\partial \theta} = \frac{\partial \mathbf{A}}{\partial \theta} \mathbf{A}^\top + \mathbf{A} \frac{\partial \mathbf{A}^\top}{\partial \theta}. \quad (\text{A.14})$$

Multiplying from the left with $\mathbf{A}^{-1}$ and from the right with $\mathbf{A}^{-\top}$ gives

$$\mathbf{A}^{-1} \frac{\partial \mathbf{P}}{\partial \theta} \mathbf{A}^{-\top} = \mathbf{A}^{-1} \frac{\partial \mathbf{A}}{\partial \theta} + \frac{\partial \mathbf{A}^\top}{\partial \theta} \mathbf{A}^{-\top}. \quad (\text{A.15})$$

Now the right-hand side is the sum of a lower triangular matrix and an upper triangular matrix with identical diagonals. Thus we can recover

$\mathbf{A}^{-1} \partial \mathbf{A} / \partial \theta$ via

$$\mathbf{A}^{-1} \frac{\partial \mathbf{A}}{\partial \theta} = \Phi \left(\mathbf{A}^{-1} \frac{\partial \mathbf{P}}{\partial \theta} \mathbf{A}^{-\top} \right), \quad (\text{A.16})$$

where the function $\Phi(\cdot)$ returns the (strictly) lower triangular part of the argument and half of the diagonal. Multiplying from the left by $\mathbf{A}$ gives the result. $\square$

A.4 Affine Stochastic Differential Equations

Many of the stochastic differential equations (SDEs) that we study in Chapter 4 are affine, and the following lemma can be used to derive several of the key results in that chapter. Note that a more rigorous derivation of these results can be obtained using Itô calculus (see, e.g., Särkkä and Solin, 2019).

Lemma A.9 (Analytical solution to affine SDE). *Suppose $\mathbf{x}(t)$ obeys the stochastic differential equation (SDE)*

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{F} \mathbf{x}(t) + \mathbf{b}(t) + \mathbf{L} \mathbf{w}(t), \quad (\text{A.17})$$

where $\mathbf{F}$ is constant, $\mathbf{b}(t)$ is a deterministic function of time, and $\mathbf{w}(t)$ is white Gaussian noise with the moments (4.4). It then holds that

$$\begin{aligned} \mathbf{x}(t + \tau) = & \exp(\mathbf{F}\tau) \mathbf{x}(t) + \int_t^{t+\tau} \exp(\mathbf{F}(t + \tau - v)) \mathbf{b}(v) dv \\ & + \int_t^{t+\tau} \exp(\mathbf{F}(t + \tau - v)) \mathbf{L} \mathbf{w}(v) dv, \end{aligned} \quad (\text{A.18})$$

where $\int_t^{t+\tau} \exp(\mathbf{F}(t + \tau - v)) \mathbf{L} \mathbf{w}(v) dv$ is a Gaussian random variable with zero mean and covariance

$$\int_0^\tau \exp(\mathbf{F}v) \mathbf{L} \mathbf{Q}^c \mathbf{L}^\top \exp(\mathbf{F}v)^\top dv. \quad (\text{A.19})$$

Consequently, $\mathbf{x}(t + \tau) \mid \mathbf{x}(t)$ is Gaussian with the moments

$$\begin{aligned} \mathbb{E}[\mathbf{x}(t + \tau) \mid \mathbf{x}(t)] = & \exp(\mathbf{F}\tau) \mathbf{x}(t) \\ & + \int_t^{t+\tau} \exp(\mathbf{F}(t + \tau - v)) \mathbf{b}(v) dv, \end{aligned} \quad (\text{A.20})$$

$$\text{Cov}[\mathbf{x}(t + \tau) \mid \mathbf{x}(t)] = \int_0^\tau \exp(\mathbf{F}v) \mathbf{L} \mathbf{Q}^c \mathbf{L}^\top \exp(\mathbf{F}v)^\top dv.$$

Furthermore, if $\mathbf{b}(t) = \mathbf{b}$ is a constant, it holds that

$$\int_t^{t+\tau} \exp(\mathbf{F}(t + \tau - v)) \mathbf{b} \, dv = \int_0^\tau \exp(\mathbf{F} v) \, dv \mathbf{b} = \left(\sum_{i=1}^{\infty} \frac{\tau^i \mathbf{F}^{i-1}}{i!} \right) \mathbf{b}, \quad (\text{A.21})$$

which further simplifies the expressions for $\mathbf{x}(t + \tau)$ and $E[\mathbf{x}(t + \tau) \mid \mathbf{x}(t)]$.

Proof To solve (A.17), we move the term $\mathbf{F} \mathbf{x}(t)$ to the left-hand side and multiply both sides of the equation with the integrating factor $\exp(-\mathbf{F} t)$, which gives

$$\exp(-\mathbf{F} t) \frac{d\mathbf{x}(t)}{dt} - \exp(-\mathbf{F} t) \mathbf{F} \mathbf{x}(t) = \exp(-\mathbf{F} t) (\mathbf{b}(t) + \mathbf{L} \mathbf{w}(t)). \quad (\text{A.22})$$

From the definition of the matrix exponential,

$$\exp(\mathbf{F} t) = \sum_{i=0}^{\infty} \frac{\mathbf{F}^i t^i}{i!} = \mathbf{I} + \mathbf{F} t + \frac{\mathbf{F}^2 t^2}{2!} + \dots, \quad (\text{A.23})$$

it follows that $\frac{d}{dt} \exp(-\mathbf{F} t) = -\mathbf{F} \exp(-\mathbf{F} t)$, and we can use this relation to identify that the left-hand side in (A.22) is $\frac{d}{dt} (\exp(-\mathbf{F} t) \mathbf{x}(t))$, which implies that

$$\frac{d}{dt} (\exp(-\mathbf{F} t) \mathbf{x}(t)) = \exp(-\mathbf{F} t) (\mathbf{b}(t) + \mathbf{L} \mathbf{w}(t)). \quad (\text{A.24})$$

Integrating (A.24) from t to $t + \tau$ therefore gives

$$\begin{aligned} \exp(-\mathbf{F}(t + \tau)) \mathbf{x}(t + \tau) - \exp(-\mathbf{F} t) \mathbf{x}(t) \\ = \int_t^{t+\tau} \exp(-\mathbf{F} v) (\mathbf{b}(t) + \mathbf{L} \mathbf{w}(v)) \, dv. \end{aligned} \quad (\text{A.25})$$

By multiplying both sides of (A.25) by $\exp(\mathbf{F}(t + \tau))$, we obtain (A.18).

Let us now analyze the two remaining integrals in (A.18) one at a time. If $\mathbf{b}(t)$ is a constant, it holds that

$$\int_t^{t+\tau} \exp(\mathbf{F}(t + \tau - v)) \mathbf{b} \, dv = \int_t^{t+\tau} \exp(\mathbf{F}(t + \tau - v)) \, dv \mathbf{b}. \quad (\text{A.26})$$

Using appropriate changes of integration variables, combined with the definition of the matrix exponential, we get

$$\begin{aligned}
 \int_t^{t+\tau} \exp(\mathbf{F}(t + \tau - v)) \, dv &= \int_0^\tau \exp(\mathbf{F} v) \, dv \\
 &= \int_0^\tau \sum_{i=0}^{\infty} \frac{\mathbf{F}^i v^i}{i!} \, dv \\
 &= \sum_{i=0}^{\infty} \frac{\mathbf{F}^i \tau^{i+1}}{(i+1)!} \\
 &= \mathbf{I}\tau + \frac{\mathbf{F}\tau^2}{2} + \frac{\mathbf{F}^2\tau^3}{3!} + \cdots,
 \end{aligned} \tag{A.27}$$

which confirms (A.21).

The last integral in (A.18) contains $\mathbf{w}(t)$, which is a Gaussian process with zero mean, and $\int_t^{t+\tau} \exp(\mathbf{F}(t + \tau - v)) \mathbf{L} \mathbf{w}(v) \, dv$ is therefore also a zero mean Gaussian random variable. Two results are central for deriving its covariance matrix. First, that the covariance matrix of a zero mean random vector is the expected value of the vector times the transpose of the vector. Second, that the definition of the Dirac delta function $\delta(\cdot)$ that appears in (4.4) is that $\int \delta(v - v_0) f(v) \, dv = f(v_0)$. Using these results we get

$$\begin{aligned}
 &\mathbf{E} \left[\int_t^{t+\tau} \exp(\mathbf{F}(t + \tau - v_1)) \mathbf{L} \mathbf{w}(v_1) \, dv_1 \right. \\
 &\quad \times \left. \int_t^{t+\tau} \mathbf{w}^\top(v_2) \mathbf{L}^\top \exp(\mathbf{F}(t + \tau - v_2))^\top \, dv_2 \right] \\
 &= \int_t^{t+\tau} \int_t^{t+\tau} \exp(\mathbf{F}(t + \tau - v_1)) \mathbf{L} \mathbf{E} [\mathbf{w}(v_1) \mathbf{w}^\top(v_2)] \mathbf{L}^\top \\
 &\quad \times \exp(\mathbf{F}(t + \tau - v_2))^\top \, dv_1 \, dv_2 \\
 &= \int_t^{t+\tau} \int_t^{t+\tau} \exp(\mathbf{F}(t + \tau - v_1)) \mathbf{L} \mathbf{Q}^c \delta(v_1 - v_2) \mathbf{L}^\top \\
 &\quad \times \exp(\mathbf{F}(t + \tau - v_2))^\top \, dv_1 \, dv_2 \\
 &= \int_t^{t+\tau} \exp(\mathbf{F}(t + \tau - v_1)) \mathbf{L} \mathbf{Q}^c \mathbf{L}^\top \exp(\mathbf{F}(t + \tau - v_1))^\top \, dv_1 \\
 &= \int_0^\tau \exp(\mathbf{F} v) \mathbf{L} \mathbf{Q}^c \mathbf{L}^\top \exp(\mathbf{F} v)^\top \, dv.
 \end{aligned} \tag{A.28}$$

The above equation confirms (A.19). Finally, (A.20) follows from the earlier results. $\square$

A.5 Time Derivative of Covariance in Theorem 4.13

In this appendix, we derive the expression for the time derivative of $\mathbf{P}(t)$ presented in Theorem 4.13. Recall that $\mathbf{P}(t) = E[(\mathbf{x}(t) - \mathbf{m}(t))(\mathbf{x}(t) - \mathbf{m}(t))^T]$ and that

$$\frac{d\mathbf{P}(t)}{dt} = \lim_{h \rightarrow 0} \frac{\mathbf{P}(t+h) - \mathbf{P}(t)}{h}. \quad (\text{A.29})$$

We therefore seek an approximation to $\mathbf{P}(t+h)$ where $o(h)$ terms are omitted, that is, we ignore terms that go to zero faster than h as $h \rightarrow 0$. First of all we have

$$\begin{aligned} \mathbf{x}(t+h) - \mathbf{x}(t) &= \int_t^{t+h} \mathbf{a}(\mathbf{m}(s)) \, ds \\ &+ \int_t^{t+h} \mathbf{A}_{\mathbf{x}}(\mathbf{m}(s)) (\mathbf{x}(s) - \mathbf{m}(s)) \, ds + \int_t^{t+h} \mathbf{L} \mathbf{w}(s) \, ds. \end{aligned} \quad (\text{A.30})$$

On the other hand,

$$\mathbf{m}(t+h) - \mathbf{m}(t) = \int_t^{t+h} \mathbf{a}(\mathbf{m}(s)) \, ds. \quad (\text{A.31})$$

We can now write

$$\begin{aligned} \mathbf{x}(t+h) - \mathbf{m}(t+h) &= \mathbf{x}(t+h) - \mathbf{m}(t+h) + \mathbf{x}(t) - \mathbf{x}(t) + \mathbf{m}(t) - \mathbf{m}(t) \\ &= [\mathbf{x}(t) - \mathbf{m}(t)] + [\mathbf{x}(t+h) - \mathbf{x}(t)] - [\mathbf{m}(t+h) - \mathbf{m}(t)] \\ &= [\mathbf{x}(t) - \mathbf{m}(t)] + \int_t^{t+h} \mathbf{A}_{\mathbf{x}}(\mathbf{m}(s)) (\mathbf{x}(s) - \mathbf{m}(s)) \, ds + \int_t^{t+h} \mathbf{L} \mathbf{w}(s) \, ds \\ &= [\mathbf{x}(t) - \mathbf{m}(t)] + \mathbf{A}_{\mathbf{x}}(\mathbf{m}(t)) (\mathbf{x}(t) - \mathbf{m}(t)) h + o(h) + \int_t^{t+h} \mathbf{L} \mathbf{w}(s) \, ds, \end{aligned} \quad (\text{A.32})$$

where we have left the integral involving white noise $\mathbf{w}(t)$ intact because $\int_t^{t+h} \mathbf{L} \mathbf{w}(s) \, ds \neq \mathbf{L} \mathbf{w}(t) h + o(h)$ (cf. Särkkä and Solin, 2019), but instead we need to explicitly use Equation (4.4). Because $\int_t^{t+h} \mathbf{L} \mathbf{w}(s) \, ds$ is

independent of the other terms and $E \left[\int_t^{t+h} \mathbf{L} \mathbf{w}(s) ds \right] = 0$, we then get

$$\begin{aligned}
 \mathbf{P}(t+h) &= E[(\mathbf{x}(t+h) - \mathbf{m}(t+h))(\mathbf{x}(t+h) - \mathbf{m}(t+h))^T] \\
 &= E[(\mathbf{x}(t) - \mathbf{m}(t))(\mathbf{x}(t) - \mathbf{m}(t))^T] \\
 &\quad + \mathbf{A}_{\mathbf{x}}(\mathbf{m}(t)) E[(\mathbf{x}(t) - \mathbf{m}(t))(\mathbf{x}(t) - \mathbf{m}(t))^T] h \\
 &\quad + E[(\mathbf{x}(t) - \mathbf{m}(t))(\mathbf{x}(t) - \mathbf{m}(t))^T] \mathbf{A}_{\mathbf{x}}^T(\mathbf{m}(t)) h \\
 &\quad + E \left[\int_t^{t+h} \mathbf{L} \mathbf{w}(s) \mathbf{w}^T(s') \mathbf{L}^T ds ds' \right] + o(h) \\
 &= \mathbf{P}(t) + \mathbf{A}_{\mathbf{x}}(\mathbf{m}(t)) \mathbf{P}(t) h + \mathbf{P}(t) \mathbf{A}_{\mathbf{x}}^T(\mathbf{m}(t)) h \\
 &\quad + \mathbf{L} \mathbf{Q}^c \mathbf{L}^T h + o(h).
 \end{aligned} \tag{A.33}$$

Rearranging and taking the limit $h \rightarrow 0$ then yields Equation (4.53).

A.6 Derivation of Mean for Bicycle Model

In this section, the aim is to derive the solution to the mean differential equation in Example 4.17. Our aim is now to solve the differential equation

$$\frac{d}{dt} \begin{pmatrix} x_1(t) \\ x_2(t) \\ s(t) \\ \theta(t) \\ \delta(t) \end{pmatrix} = \begin{pmatrix} s(t) \cos(\theta(t)) \\ s(t) \sin(\theta(t)) \\ 0 \\ s(t)/r(t) \\ 0 \end{pmatrix} \tag{A.34}$$

where $r(t) = L/\tan(\delta(t))$ for $\mathbf{x}(t) = (x_1(t), x_2(t), s(t), \theta(t), \delta(t))$ given the initial condition $\mathbf{x}(t_{k-1}) = \mathbf{x}_{k-1}$. It holds that $s(t) = s_{k-1}$, $\delta(t) = \delta_{k-1}$, and thus $r(t) = r_{k-1} = L/\tan(\delta_{k-1})$, which implies that $\theta(t) = \theta_{k-1} + (t - t_{k-1}) s_{k-1}/r_{k-1}$. We can now solve for

$$\begin{aligned}
 x_1(t) &= x_{1,k-1} + \int_{t_{k-1}}^t s(v) \cos(\theta(v)) dv \\
 &= x_{1,k-1} + \int_0^{t-t_{k-1}} s_{k-1} \cos(\theta_{k-1} + v s_{k-1}/r_{k-1}) dv \\
 &= x_{1,k-1} + r_{k-1} (\sin(\theta_{k-1} + (t - t_{k-1}) s_{k-1}/r_{k-1}) - \sin(\theta_{k-1})).
 \end{aligned} \tag{A.35}$$

Using a similar derivation for $x_2(t)$, we finally get

$$\begin{pmatrix} x_1(t) \\ x_2(t) \\ s(t) \\ \theta(t) \\ \delta(t) \end{pmatrix} = \begin{pmatrix} x_{1,k-1} + r_{k-1} (\sin(\theta_{k-1} + \beta(t - t_{k-1})) - \sin(\theta_{k-1})) \\ x_{2,k-1} + r_{k-1} (\cos(\theta_{k-1}) - \cos(\theta_{k-1} + \beta(t - t_{k-1}))) \\ s_{k-1} \\ \theta_{k-1} + \beta(t - t_{k-1}) \\ \delta_{k-1} \end{pmatrix}, \quad (\text{A.36})$$

where for brevity, we have defined $\beta(t) = t s_{k-1}/r_{k-1}$. When the above solution is evaluated at $t = t_k$, the corresponding discretized model then becomes Equation (4.66).

A.7 Mean Discretization for the Polar Coordinated Turn Model

Let us now derive the solution to the differential equation (4.56) that corresponds to the mean discretization of the polar coordinated turn model. The state is given as $\mathbf{x}(t) = (x_1(t), x_2(t), s(t), \varphi(t), \omega(t))$, and the initial condition is $\mathbf{x}(t_{k-1}) = \mathbf{x}_{k-1}$. Because the time derivatives of $s(t)$ and $\omega(t)$ are zero, it holds that

$$s(t) = s_{k-1}, \quad \omega(t) = \omega_{k-1}. \quad (\text{A.37})$$

Because $\omega(t)$ is the time derivative of $\varphi(t)$, it follows that

$$\varphi(t) = \varphi(t_{k-1}) + (t - t_{k-1}) \omega(t_{k-1}). \quad (\text{A.38})$$

Given (A.37) and (A.38), we get

$$\begin{aligned} x_1(t) &= x_{1,k-1} + \int_{t_{k-1}}^t s(v) \cos(\varphi(v)) \, dv \\ &= x_{1,k-1} + \int_{t_{k-1}}^t s_{k-1} \cos(\varphi_{k-1} + (v - t_{k-1}) \omega_{k-1}) \, dv \\ &= x_{1,k-1} + \frac{s_{k-1}}{\omega_{k-1}} (\sin(\varphi_{k-1} + (t - t_{k-1}) \omega_{k-1}) - \sin(\varphi_{k-1})) \\ &= x_{1,k-1} + \frac{2s_{k-1}}{\omega(t_{k-1})} \sin\left(\frac{(t - t_{k-1}) \omega_{k-1}}{2}\right) \\ &\quad \times \cos\left(\varphi_{k-1} + \frac{(t - t_{k-1}) \omega_{k-1}}{2}\right) \end{aligned} \quad (\text{A.39})$$

and, using similar calculations,

$$x_2(t) = x_{2,k-1} + \frac{2s_{k-1}}{\omega_{k-1}} \sin\left(\frac{(t - t_{k-1})\omega(t_{k-1})}{2}\right) \times \sin\left(\varphi_{k-1} + \frac{(t - t_{k-1})\omega(t_{k-1})}{2}\right). \quad (\text{A.40})$$

The final discretization is obtained by evaluation at $t = t_k$.

A.8 Approximating $\mathbf{Q}_{k-1}$ in the Polar Coordinated Turn Model

Let us now derive a constant gradient-based covariance approximation as given by Algorithm 4.19 for the polar coordinated turn model, whose mean discretization was considered in Example 4.15. The derivation of the mean discretization is given above in Section A.7. The covariance expression is thus given as

$$\mathbf{Q}_{k-1} = \int_0^{\Delta t_{k-1}} \exp(\mathbf{A}_{\mathbf{x}}(\mathbf{x}_{k-1})v) \mathbf{L} \mathbf{Q}^c \mathbf{L}^T \exp(\mathbf{A}_{\mathbf{x}}(\mathbf{x}_{k-1})v)^T dv. \quad (\text{A.41})$$

For brevity, we write

$$\mathbf{x}_{k-1} = (x_1 \quad x_2 \quad s \quad \varphi \quad \omega)^T. \quad (\text{A.42})$$

It follows from (4.56) that

$$\mathbf{A}_{\mathbf{x}}(\mathbf{x}_{k-1}) = \begin{pmatrix} 0 & 0 & \cos(\varphi) & -s \sin(\varphi) & 0 \\ 0 & 0 & \sin(\varphi) & s \cos(\varphi) & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad (\text{A.43})$$

and it is easy to verify that

$$(\mathbf{A}_{\mathbf{x}}(\mathbf{x}_{k-1}))^2 = \begin{pmatrix} 0 & 0 & 0 & 0 & -s \sin(\varphi) \\ 0 & 0 & 0 & 0 & s \cos(\varphi) \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad (\text{A.44})$$

whereas $(\mathbf{A}_{\mathbf{x}}(\mathbf{x}_{k-1}))^n = \mathbf{0}$ for $n \geq 3$. The definition of the matrix exponential therefore gives

$$\begin{aligned} & \exp(\mathbf{A}_{\mathbf{x}}(\mathbf{x}_{k-1}) v) \\ &= \mathbf{I} + v \mathbf{A}_{\mathbf{x}}(\mathbf{x}_{k-1}) + \frac{v^2 (\mathbf{A}_{\mathbf{x}}(\mathbf{x}_{k-1}))^2}{2} \\ &= \begin{pmatrix} 1 & 0 & v \cos(\varphi) & -vs \sin(\varphi) & -\frac{v^2}{2}s \sin(\varphi) \\ 0 & 1 & v \sin(\varphi) & vs \cos(\varphi) & \frac{v^2}{2}s \cos(\varphi) \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & v \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \end{aligned} \quad (\text{A.45})$$

and from the expression for $\mathbf{L}$ in (4.43) we now obtain

$$\exp(\mathbf{A}_{\mathbf{x}}(\mathbf{x}_{k-1}) v) \mathbf{L} = \begin{pmatrix} v \cos(\varphi) & -\frac{v^2}{2}s \sin(\varphi) \\ v \sin(\varphi) & \frac{v^2}{2}s \cos(\varphi) \\ 1 & 0 \\ 0 & v \\ 0 & 1 \end{pmatrix}. \quad (\text{A.46})$$

For simplicity, we assume that $\mathbf{Q}^c = \text{diag}(q_1, q_2)$. To save space, we use the shorthand notations $\text{si} = \sin(\varphi)$, $\text{co} = \cos(\varphi)$ and $\text{cs} = \sin(\varphi) \cos(\varphi)$ and we only express the lower triangle of the following symmetric matrices. Equation (A.46) yields

$$\begin{aligned} & \exp(\mathbf{A}_{\mathbf{x}}(\mathbf{x}_{k-1}) v) \mathbf{L} \mathbf{Q}^c \mathbf{L}^\top \exp(\mathbf{A}_{\mathbf{x}}(\mathbf{x}_{k-1}) v)^\top \\ &= \begin{pmatrix} q_1 v^2 \text{co}^2 + q_2 \frac{v^4}{4} s^2 \text{si}^2 & & & & \\ q_1 v^2 \text{cs} - q_2 \frac{v^4}{4} s^2 \text{cs} & q_1 v^2 \text{si}^2 + q_2 \frac{v^4}{4} s^2 \text{co}^2 & & & \\ q_1 v \text{co} & q_1 v \text{si} & q_1 & & \\ -q_2 s \frac{v^3}{2} \text{si} & q_2 s \frac{v^3}{2} \text{co} & 0 & q_2 v^2 & \\ -q_2 s \frac{v^2}{2} \text{si} & q_2 s \frac{v^2}{2} \text{co} & 0 & q_2 v & q_2 \end{pmatrix}. \end{aligned} \quad (\text{A.47})$$

Finally, we integrate over v to obtain

$$\begin{aligned} & \mathbf{P}(\tau + t_{k-1}) \\ &= \begin{pmatrix} q_1 \frac{\tau^3}{3} \text{co}^2 + q_2 \frac{\tau^5}{20} s^2 \text{si}^2 & q_1 \frac{\tau^3}{3} \text{si}^2 + q_2 \frac{\tau^5}{20} s^2 \text{co}^2 & & & \\ q_1 \frac{\tau^3}{3} \text{cs} - q_2 \frac{\tau^5}{20} s^2 \text{cs} & & & & \\ q_1 \frac{\tau^2}{2} \text{co} & q_1 \frac{\tau^2}{2} \text{si} & q_1 \tau & & \\ -q_2 s \frac{\tau^4}{8} \text{si} & q_2 s \frac{\tau^4}{8} \text{co} & 0 & q_2 \frac{\tau^3}{3} & \\ -q_2 s \frac{\tau^3}{6} \text{si} & q_2 s \frac{\tau^3}{6} \text{co} & 0 & q_2 \frac{\tau^2}{2} & q_2 \tau \end{pmatrix}, \end{aligned} \quad (\text{A.48})$$

which gives $\mathbf{Q}_{k-1}$ by setting $\tau = \Delta t_{k-1}$.

A.9 Conditional Moments Used in SLR

We can use statistical linear regression (SLR) to approximate the dynamic and measurement models in both filtering and smoothing. Theorems 9.12 and 9.16, combined with Corollaries 9.14, 9.17, and 9.18, provide five different expressions for moments that we need to compute to perform SLR; one expression for the additive noise models in (9.1) and two expressions each for the non-additive noise models in (9.9) and for the conditional distribution models in (9.10). In this section, we present the explicit forms that these equations take for the dynamic and measurement models, respectively.

Using SLR, the dynamic model is approximated as

$$\mathbf{x}_k \simeq \mathbf{A}_{k-1} \mathbf{x}_{k-1} + \mathbf{a}_{k-1} + \mathbf{e}_{k-1}, \quad (\text{A.49})$$

where $\mathbf{e}_{k-1} \sim \mathbf{N}(\mathbf{0}, \mathbf{\Lambda}_{k-1})$. If¹ $\mathbf{x}_{k-1} \sim \mathbf{N}(\tilde{\mathbf{m}}_{k-1}, \tilde{\mathbf{P}}_{k-1})$, it holds that

$$\begin{aligned} \mathbf{A}_{k-1} &= (\mathbf{P}_k^{\text{xx}})^\top \tilde{\mathbf{P}}_{k-1}^{-1}, \\ \mathbf{a}_{k-1} &= \boldsymbol{\mu}_k^- - \mathbf{A}_{k-1} \tilde{\mathbf{m}}_{k-1}, \\ \mathbf{\Lambda}_{k-1} &= \mathbf{P}_k^{\text{x}} - \mathbf{A}_{k-1} \tilde{\mathbf{P}}_{k-1} \mathbf{A}_{k-1}^\top, \end{aligned} \quad (\text{A.50})$$

where Theorem 9.16 and Corollaries 9.14, 9.17, and 9.18 provide the following five expressions for $\boldsymbol{\mu}_k^- = \mathbf{E}[\mathbf{x}_k]$, $\mathbf{P}_k^{\text{xx}} = \text{Cov}[\mathbf{x}_{k-1}, \mathbf{x}_k]$, and $\mathbf{P}_k^{\text{x}} =$

¹ We use tildes to indicate that the equations are valid for any mean and covariance, but note that the choice of these moments influences the resulting SLR approximation.

$\text{Cov}[\mathbf{x}_k]$. For additive noise models, Corollary 9.14 implies that

$$\begin{aligned}\mu_k^- &= \int \mathbf{f}(\mathbf{x}_{k-1}) \mathbf{N}(\mathbf{x}_{k-1} \mid \tilde{\mathbf{m}}_{k-1}, \tilde{\mathbf{P}}_{k-1}) d\mathbf{x}_{k-1}, \\ \mathbf{P}_k^{\text{xx}} &= \int (\mathbf{x}_{k-1} - \tilde{\mathbf{m}}_{k-1}) (\mathbf{f}(\mathbf{x}_{k-1}) - \mu_k^-)^\top \mathbf{N}(\mathbf{x}_{k-1} \mid \tilde{\mathbf{m}}_{k-1}, \tilde{\mathbf{P}}_{k-1}) d\mathbf{x}_{k-1}, \\ \mathbf{P}_k^{\text{x}} &= \int (\mathbf{f}(\mathbf{x}_{k-1}) - \mu_k^-) (\mathbf{f}(\mathbf{x}_{k-1}) - \mu_k^-)^\top \mathbf{N}(\mathbf{x}_{k-1} \mid \tilde{\mathbf{m}}_{k-1}, \tilde{\mathbf{P}}_{k-1}) d\mathbf{x}_{k-1} \\ &\quad + \mathbf{Q}_{k-1};\end{aligned}\tag{A.51}$$

for non-additive noise models, Equations (9.50) in Corollary 9.17 give

$$\begin{aligned}\mu_k^- &= \int \mathbf{f}(\mathbf{x}_{k-1}, \mathbf{q}_{k-1}) \\ &\quad \times \mathbf{N}(\mathbf{x}_{k-1} \mid \tilde{\mathbf{m}}_{k-1}, \tilde{\mathbf{P}}_{k-1}) \mathbf{N}(\mathbf{q}_{k-1} \mid \mathbf{0}, \mathbf{Q}_{k-1}) d\mathbf{x}_{k-1} d\mathbf{q}_{k-1}, \\ \mathbf{P}_k^{\text{xx}} &= \int (\mathbf{x}_{k-1} - \tilde{\mathbf{m}}_{k-1}) (\mathbf{f}(\mathbf{x}_{k-1}, \mathbf{q}_{k-1}) - \mu_k^-)^\top \\ &\quad \times \mathbf{N}(\mathbf{x}_{k-1} \mid \tilde{\mathbf{m}}_{k-1}, \tilde{\mathbf{P}}_{k-1}) \mathbf{N}(\mathbf{q}_{k-1} \mid \mathbf{0}, \mathbf{Q}_{k-1}) d\mathbf{x}_{k-1} d\mathbf{q}_{k-1}, \\ \mathbf{P}_k^{\text{x}} &= \int (\mathbf{f}(\mathbf{x}_{k-1}, \mathbf{q}_{k-1}) - \mu_k^-) (\mathbf{f}(\mathbf{x}_{k-1}, \mathbf{q}_{k-1}) - \mu_k^-)^\top \\ &\quad \times \mathbf{N}(\mathbf{x}_{k-1} \mid \tilde{\mathbf{m}}_{k-1}, \tilde{\mathbf{P}}_{k-1}) \mathbf{N}(\mathbf{q}_{k-1} \mid \mathbf{0}, \mathbf{Q}_{k-1}) d\mathbf{x}_{k-1} d\mathbf{q}_{k-1};\end{aligned}\tag{A.52}$$

and for conditional distribution models, Equations (9.53) in Corollary 9.18 give

$$\begin{aligned}\mu_k^- &= \int \mathbf{x}_k p(\mathbf{x}_k \mid \mathbf{x}_{k-1}) \mathbf{N}(\mathbf{x}_{k-1} \mid \tilde{\mathbf{m}}_{k-1}, \tilde{\mathbf{P}}_{k-1}) d\mathbf{x}_k d\mathbf{x}_{k-1}, \\ \mathbf{P}_k^{\text{xx}} &= \int (\mathbf{x}_{k-1} - \tilde{\mathbf{m}}_{k-1}) (\mathbf{x}_k - \mu_k^-)^\top \\ &\quad \times p(\mathbf{x}_k \mid \mathbf{x}_{k-1}) \mathbf{N}(\mathbf{x}_{k-1} \mid \tilde{\mathbf{m}}_{k-1}, \tilde{\mathbf{P}}_{k-1}) d\mathbf{x}_k d\mathbf{x}_{k-1}, \\ \mathbf{P}_k^{\text{x}} &= \int (\mathbf{x}_k - \mu_k^-) (\mathbf{x}_k - \mu_k^-)^\top \\ &\quad \times p(\mathbf{x}_k \mid \mathbf{x}_{k-1}) \mathbf{N}(\mathbf{x}_{k-1} \mid \tilde{\mathbf{m}}_{k-1}, \tilde{\mathbf{P}}_{k-1}) d\mathbf{x}_k d\mathbf{x}_{k-1}.\end{aligned}\tag{A.53}$$

Theorem 9.16 also implies that we can use the conditional moments formulation to obtain alternative expressions for μ_k^- , $\mathbf{P}_k^{\text{xx}}$, and $\mathbf{P}_k^{\text{x}}$. Introducing

$\mu_k^-(\mathbf{x}_{k-1}) = E[\mathbf{x}_k | \mathbf{x}_{k-1}]$ and $\mathbf{P}_k^x(\mathbf{x}_{k-1}) = \text{Cov}[\mathbf{x}_k | \mathbf{x}_{k-1}]$, it holds that

$$\begin{aligned}\mu_k^- &= \int \mu_k^-(\mathbf{x}_{k-1}) N(\mathbf{x}_{k-1} | \tilde{\mathbf{m}}_{k-1}, \tilde{\mathbf{P}}_{k-1}) d\mathbf{x}_{k-1}, \\ \mathbf{P}_k^{xx} &= \int (\mathbf{x}_{k-1} - \tilde{\mathbf{m}}_{k-1}) (\mu_k^-(\mathbf{x}_{k-1}) - \mu_k^-)^\top \\ &\quad \times N(\mathbf{x}_{k-1} | \tilde{\mathbf{m}}_{k-1}, \tilde{\mathbf{P}}_{k-1}) d\mathbf{x}_{k-1}, \\ \mathbf{P}_k^x &= \int (\mu_k^-(\mathbf{x}_{k-1}) - \mu_k^-) (\mu_k^-(\mathbf{x}_{k-1}) - \mu_k^-)^\top \\ &\quad \times N(\mathbf{x}_{k-1} | \tilde{\mathbf{m}}_{k-1}, \tilde{\mathbf{P}}_{k-1}) d\mathbf{x}_{k-1} \\ &\quad + \int \mathbf{P}_k^x(\mathbf{x}_{k-1}) N(\mathbf{x}_{k-1} | \tilde{\mathbf{m}}_{k-1}, \tilde{\mathbf{P}}_{k-1}) d\mathbf{x}_{k-1}.\end{aligned}\tag{A.54}$$

For additive noise models, $\mu_k^-(\mathbf{x}_{k-1}) = \mathbf{f}(\mathbf{x}_{k-1})$ and $\mathbf{P}_k^x(\mathbf{x}_{k-1}) = \mathbf{Q}_{k-1}$, and (A.54) simplifies to (A.51). For non-additive noise models, Equations (9.51) in Corollary 9.17 imply that

$$\begin{aligned}\mu_k^-(\mathbf{x}_{k-1}) &= \int \mathbf{f}(\mathbf{x}_{k-1}, \mathbf{q}_{k-1}) N(\mathbf{q}_{k-1} | \mathbf{0}, \mathbf{Q}_{k-1}) d\mathbf{q}_{k-1}, \\ \mathbf{P}_k^x(\mathbf{x}_{k-1}) &= \int (\mathbf{f}(\mathbf{x}_{k-1}, \mathbf{q}_{k-1}) - \mu_k^-(\mathbf{x}_{k-1})) \\ &\quad \times (\mathbf{f}(\mathbf{x}_{k-1}, \mathbf{q}_{k-1}) - \mu_k^-(\mathbf{x}_{k-1}))^\top N(\mathbf{q}_{k-1} | \mathbf{0}, \mathbf{Q}_{k-1}) d\mathbf{q}_{k-1}.\end{aligned}\tag{A.55}$$

Finally, for conditional distribution models, Equations (9.54) in Corollary 9.18 give

$$\begin{aligned}\mu_k^-(\mathbf{x}_{k-1}) &= \int \mathbf{x}_k p(\mathbf{x}_k | \mathbf{x}_{k-1}) d\mathbf{x}_k, \\ \mathbf{P}_k^x(\mathbf{x}_{k-1}) &= \int (\mathbf{x}_k - \mu_k^-(\mathbf{x}_{k-1})) (\mathbf{x}_k - \mu_k^-(\mathbf{x}_{k-1}))^\top p(\mathbf{x}_k | \mathbf{x}_{k-1}) d\mathbf{x}_k.\end{aligned}\tag{A.56}$$

In total, the above equations present five different alternatives to compute the moments μ_k^- , $\mathbf{P}_k^{xx}$, and $\mathbf{P}_k^x$ that we need to approximate the dynamic model using SLR.

Similarly, we can use SLR to approximate the measurement model as

$$\mathbf{y}_k \simeq \mathbf{H}_k \mathbf{x}_k + \mathbf{b}_k + \mathbf{v}_k,\tag{A.57}$$

where $\mathbf{v}_k \sim N(\mathbf{0}, \mathbf{\Omega}_k)$. If $\mathbf{x}_k \sim N(\tilde{\mathbf{m}}_k, \tilde{\mathbf{P}}_k)$, it holds that

$$\begin{aligned}\mathbf{H}_k &= (\mathbf{P}_k^{\text{xy}})^\top \tilde{\mathbf{P}}_k^{-1}, \\ \mathbf{b}_k &= \boldsymbol{\mu}_k^+ - \mathbf{H}_k \tilde{\mathbf{m}}_k, \\ \mathbf{\Omega}_k &= \mathbf{P}_k^y - \mathbf{H}_k \tilde{\mathbf{P}}_k \mathbf{H}_k^\top,\end{aligned}\tag{A.58}$$

where we can obtain five different expressions for $\boldsymbol{\mu}_k^+ = E[\mathbf{y}_k]$, $\mathbf{P}_k^{\text{xy}} = \text{Cov}[\mathbf{x}_k, \mathbf{y}_k]$, and $\mathbf{P}_k^y = \text{Cov}[\mathbf{y}_k]$ from Theorem 9.16 and Corollaries 9.14, 9.17, and 9.18. For additive noise models, Corollary 9.14 implies that

$$\begin{aligned}\boldsymbol{\mu}_k^+ &= \int \mathbf{h}(\mathbf{x}_k) N(\mathbf{x}_k \mid \tilde{\mathbf{m}}_k, \tilde{\mathbf{P}}_k) d\mathbf{x}_k, \\ \mathbf{P}_k^{\text{xy}} &= \int (\mathbf{x}_k - \tilde{\mathbf{m}}_k) (\mathbf{h}(\mathbf{x}_k) - \boldsymbol{\mu}_k^+)^\top N(\mathbf{x}_k \mid \tilde{\mathbf{m}}_k, \tilde{\mathbf{P}}_k) d\mathbf{x}_k, \\ \mathbf{P}_k^y &= \int (\mathbf{h}(\mathbf{x}_k) - \boldsymbol{\mu}_k^+) (\mathbf{h}(\mathbf{x}_k) - \boldsymbol{\mu}_k^+)^\top N(\mathbf{x}_k \mid \tilde{\mathbf{m}}_k, \tilde{\mathbf{P}}_k) d\mathbf{x}_k + \mathbf{R}_k;\end{aligned}\tag{A.59}$$

for non-additive noise models, Equations (9.50) in Corollary 9.17 give

$$\begin{aligned}\boldsymbol{\mu}_k^+ &= \int \mathbf{h}(\mathbf{x}_k, \mathbf{r}_k) N(\mathbf{x}_k \mid \tilde{\mathbf{m}}_k, \tilde{\mathbf{P}}_k) N(\mathbf{r}_k \mid \mathbf{0}, \mathbf{R}_k) d\mathbf{x}_k d\mathbf{r}_k, \\ \mathbf{P}_k^{\text{xy}} &= \int (\mathbf{x}_k - \tilde{\mathbf{m}}_k) (\mathbf{h}(\mathbf{x}_k, \mathbf{r}_k) - \boldsymbol{\mu}_k^+)^\top \\ &\quad \times N(\mathbf{x}_k \mid \tilde{\mathbf{m}}_k, \tilde{\mathbf{P}}_k) N(\mathbf{r}_k \mid \mathbf{0}, \mathbf{R}_k) d\mathbf{x}_k d\mathbf{r}_k, \\ \mathbf{P}_k^y &= \int (\mathbf{h}(\mathbf{x}_k, \mathbf{r}_k) - \boldsymbol{\mu}_k^+) (\mathbf{h}(\mathbf{x}_k, \mathbf{r}_k) - \boldsymbol{\mu}_k^+)^\top \\ &\quad \times N(\mathbf{x}_k \mid \tilde{\mathbf{m}}_k, \tilde{\mathbf{P}}_k) N(\mathbf{r}_k \mid \mathbf{0}, \mathbf{R}_k) d\mathbf{x}_k d\mathbf{r}_k;\end{aligned}\tag{A.60}$$

and for conditional distribution models, Equations (9.53) in Corollary 9.18 give

$$\begin{aligned}\boldsymbol{\mu}_k^+ &= \int \mathbf{y}_k p(\mathbf{y}_k \mid \mathbf{x}_k) N(\mathbf{x}_k \mid \tilde{\mathbf{m}}_k, \tilde{\mathbf{P}}_k) d\mathbf{y}_k d\mathbf{x}_k, \\ \mathbf{P}_k^{\text{xy}} &= \int (\mathbf{x}_k - \tilde{\mathbf{m}}_k) (\mathbf{y}_k - \boldsymbol{\mu}_k^+)^\top p(\mathbf{y}_k \mid \mathbf{x}_k) N(\mathbf{x}_k \mid \tilde{\mathbf{m}}_k, \tilde{\mathbf{P}}_k) d\mathbf{y}_k d\mathbf{x}_k, \\ \mathbf{P}_k^y &= \int (\mathbf{y}_k - \boldsymbol{\mu}_k^+) (\mathbf{y}_k - \boldsymbol{\mu}_k^+)^\top p(\mathbf{y}_k \mid \mathbf{x}_k) N(\mathbf{x}_k \mid \tilde{\mathbf{m}}_k, \tilde{\mathbf{P}}_k) d\mathbf{y}_k d\mathbf{x}_k.\end{aligned}\tag{A.61}$$

The conditional moments formulation provides two more expressions for the required moments. Introducing $\boldsymbol{\mu}_k^+(\mathbf{x}_k) = E[\mathbf{y}_k \mid \mathbf{x}_k]$ and

$\mathbf{P}_k^y(\mathbf{x}_k) = \text{Cov}[\mathbf{y}_k \mid \mathbf{x}_k]$, Theorem 9.16 gives that

$$\begin{aligned}\boldsymbol{\mu}_k^+ &= \int \boldsymbol{\mu}_k^+(\mathbf{x}_k) \mathbf{N}(\mathbf{x}_k \mid \tilde{\mathbf{m}}_k, \tilde{\mathbf{P}}_k) d\mathbf{x}_k, \\ \mathbf{P}_k^{xy} &= \int (\mathbf{x}_k - \tilde{\mathbf{m}}_k) (\boldsymbol{\mu}_k^+(\mathbf{x}_k) - \boldsymbol{\mu}_k^+)^T \mathbf{N}(\mathbf{x}_k \mid \tilde{\mathbf{m}}_k, \tilde{\mathbf{P}}_k) d\mathbf{x}_k, \\ \mathbf{P}_k^y &= \int (\boldsymbol{\mu}_k^+(\mathbf{x}_k) - \boldsymbol{\mu}_k^+) (\boldsymbol{\mu}_k^+(\mathbf{x}_k) - \boldsymbol{\mu}_k^+)^T \mathbf{N}(\mathbf{x}_k \mid \tilde{\mathbf{m}}_k, \tilde{\mathbf{P}}_k) d\mathbf{x}_k \\ &\quad + \int \mathbf{P}_k^y(\mathbf{x}_k) \mathbf{N}(\mathbf{x}_k \mid \tilde{\mathbf{m}}_k, \tilde{\mathbf{P}}_k) d\mathbf{x}_k.\end{aligned}\tag{A.62}$$

For additive noise models, $\boldsymbol{\mu}_k^+(\mathbf{x}_k) = \mathbf{h}(\mathbf{x}_k)$ and $\mathbf{P}_k^y(\mathbf{x}_k) = \mathbf{R}_k$, which means that (A.62) simplifies to (A.59). For non-additive noise models, Equations (9.51) in Corollary 9.17 imply that

$$\begin{aligned}\boldsymbol{\mu}_k^+(\mathbf{x}_k) &= \int \mathbf{h}(\mathbf{x}_k, \mathbf{r}_k) \mathbf{N}(\mathbf{r}_k \mid \mathbf{0}, \mathbf{R}_k) d\mathbf{r}_k, \\ \mathbf{P}_k^y(\mathbf{x}_k) &= \int (\mathbf{h}(\mathbf{x}_k, \mathbf{r}_k) - \boldsymbol{\mu}_k^+(\mathbf{x}_k)) (\mathbf{h}(\mathbf{x}_k, \mathbf{r}_k) - \boldsymbol{\mu}_k^+(\mathbf{x}_k))^T \\ &\quad \times \mathbf{N}(\mathbf{r}_k \mid \mathbf{0}, \mathbf{R}_k) d\mathbf{r}_k;\end{aligned}\tag{A.63}$$

and, finally, for conditional distribution models, Equations (9.54) in Corollary 9.18 give

$$\begin{aligned}\boldsymbol{\mu}_k^+(\mathbf{x}_k) &= \int \mathbf{y}_k p(\mathbf{y}_k \mid \mathbf{x}_k) d\mathbf{y}_k, \\ \mathbf{P}_k^y(\mathbf{x}_k) &= \int (\mathbf{y}_k - \boldsymbol{\mu}_k^+(\mathbf{x}_k)) (\mathbf{y}_k - \boldsymbol{\mu}_k^+(\mathbf{x}_k))^T p(\mathbf{y}_k \mid \mathbf{x}_k) d\mathbf{y}_k.\end{aligned}\tag{A.64}$$

We have now presented five different alternatives for computing the moments $\boldsymbol{\mu}_k^+$, $\mathbf{P}_k^{xy}$, and $\mathbf{P}_k^y$, which are required to approximate the measurement model using SLR.

A.10 Parameter Derivatives for the Kalman Filter

Theorem 16.9 gives the energy function (i.e., the negative logarithm of the unnormalized posterior distribution) of the parameters for the following linear Gaussian model:

$$\begin{aligned}\mathbf{x}_k &= \mathbf{A}(\boldsymbol{\theta}) \mathbf{x}_{k-1} + \mathbf{q}_{k-1}, \\ \mathbf{y}_k &= \mathbf{H}(\boldsymbol{\theta}) \mathbf{x}_k + \mathbf{r}_k,\end{aligned}\tag{A.65}$$

where $\mathbf{q}_{k-1} \sim N(\mathbf{0}, \mathbf{Q}(\boldsymbol{\theta}))$, $\mathbf{r}_k \sim N(\mathbf{0}, \mathbf{R}(\boldsymbol{\theta}))$, and $\mathbf{x}_0 \sim N(\mathbf{m}_0(\boldsymbol{\theta}), \mathbf{P}_0(\boldsymbol{\theta}))$. The parameters' derivatives, which are needed, for example, for implementing a gradient-based optimization method for finding ML or MAP estimates, can be evaluated via the following *sensitivity equations* (Gupta and Mehra, 1974), which can be derived by termwise differentiation of the energy function and Kalman filter equations in Theorem 16.9.

Theorem A.10 (Energy function derivative for linear Gaussian model I). *The derivative of the energy function given in Theorem 16.9 can be computed via the following recursion along with Kalman filtering:*

$$\begin{aligned} \frac{\partial \varphi_k(\boldsymbol{\theta})}{\partial \theta_i} &= \frac{\partial \varphi_{k-1}(\boldsymbol{\theta})}{\partial \theta_i} + \frac{1}{2} \operatorname{tr} \left(\mathbf{S}_k^{-1}(\boldsymbol{\theta}) \frac{\partial \mathbf{S}_k(\boldsymbol{\theta})}{\partial \theta_i} \right) + \mathbf{v}_k^\top(\boldsymbol{\theta}) \mathbf{S}_k^{-1}(\boldsymbol{\theta}) \frac{\partial \mathbf{v}_k(\boldsymbol{\theta})}{\partial \theta_i} \\ &\quad - \frac{1}{2} \mathbf{v}_k^\top(\boldsymbol{\theta}) \mathbf{S}_k^{-1}(\boldsymbol{\theta}) \frac{\partial \mathbf{S}_k(\boldsymbol{\theta})}{\partial \theta_i} \mathbf{S}_k^{-1}(\boldsymbol{\theta}) \mathbf{v}_k(\boldsymbol{\theta}), \end{aligned} \quad (\text{A.66})$$

where on the Kalman filter prediction step, we compute

$$\begin{aligned} \frac{\partial \mathbf{m}_k^-(\boldsymbol{\theta})}{\partial \theta_i} &= \frac{\partial \mathbf{A}(\boldsymbol{\theta})}{\partial \theta_i} \mathbf{m}_{k-1}(\boldsymbol{\theta}) + \mathbf{A}(\boldsymbol{\theta}) \frac{\partial \mathbf{m}_{k-1}(\boldsymbol{\theta})}{\partial \theta_i}, \\ \frac{\partial \mathbf{P}_k^-(\boldsymbol{\theta})}{\partial \theta_i} &= \frac{\partial \mathbf{A}(\boldsymbol{\theta})}{\partial \theta_i} \mathbf{P}_{k-1}(\boldsymbol{\theta}) \mathbf{A}^\top(\boldsymbol{\theta}) + \mathbf{A}(\boldsymbol{\theta}) \frac{\partial \mathbf{P}_{k-1}(\boldsymbol{\theta})}{\partial \theta_i} \mathbf{A}^\top(\boldsymbol{\theta}) \\ &\quad + \mathbf{A}(\boldsymbol{\theta}) \mathbf{P}_{k-1}(\boldsymbol{\theta}) \frac{\partial \mathbf{A}^\top(\boldsymbol{\theta})}{\partial \theta_i} + \frac{\partial \mathbf{Q}(\boldsymbol{\theta})}{\partial \theta_i}, \end{aligned} \quad (\text{A.67})$$

and on the Kalman filter update step, we compute

$$\begin{aligned}
 \frac{\partial \mathbf{v}_k(\boldsymbol{\theta})}{\partial \theta_i} &= -\frac{\partial \mathbf{H}(\boldsymbol{\theta})}{\partial \theta_i} \mathbf{m}_k^-(\boldsymbol{\theta}) - \mathbf{H}(\boldsymbol{\theta}) \frac{\partial \mathbf{m}_k^-(\boldsymbol{\theta})}{\partial \theta_i}, \\
 \frac{\partial \mathbf{S}_k(\boldsymbol{\theta})}{\partial \theta_i} &= \frac{\partial \mathbf{H}(\boldsymbol{\theta})}{\partial \theta_i} \mathbf{P}_k^-(\boldsymbol{\theta}) \mathbf{H}^\top(\boldsymbol{\theta}) + \mathbf{H}(\boldsymbol{\theta}) \frac{\partial \mathbf{P}_k^-(\boldsymbol{\theta})}{\partial \theta_i} \mathbf{H}^\top(\boldsymbol{\theta}) \\
 &\quad + \mathbf{H}(\boldsymbol{\theta}) \mathbf{P}_k^-(\boldsymbol{\theta}) \frac{\partial \mathbf{H}^\top(\boldsymbol{\theta})}{\partial \theta_i} + \frac{\partial \mathbf{R}(\boldsymbol{\theta})}{\partial \theta_i}, \\
 \frac{\partial \mathbf{K}_k(\boldsymbol{\theta})}{\partial \theta_i} &= \frac{\partial \mathbf{P}_k^-(\boldsymbol{\theta})}{\partial \theta_i} \mathbf{H}^\top(\boldsymbol{\theta}) \mathbf{S}_k^{-1}(\boldsymbol{\theta}) + \mathbf{P}_k^-(\boldsymbol{\theta}) \frac{\partial \mathbf{H}^\top(\boldsymbol{\theta})}{\partial \theta_i} \mathbf{S}_k^{-1}(\boldsymbol{\theta}) \\
 &\quad - \mathbf{P}_k^-(\boldsymbol{\theta}) \mathbf{H}^\top(\boldsymbol{\theta}) \mathbf{S}_k^{-1}(\boldsymbol{\theta}) \frac{\partial \mathbf{S}_k(\boldsymbol{\theta})}{\partial \theta_i} \mathbf{S}_k^{-1}(\boldsymbol{\theta}), \\
 \frac{\partial \mathbf{m}_k(\boldsymbol{\theta})}{\partial \theta_i} &= \frac{\partial \mathbf{m}_k^-(\boldsymbol{\theta})}{\partial \theta_i} + \frac{\partial \mathbf{K}_k(\boldsymbol{\theta})}{\partial \theta_i} \mathbf{v}_k(\boldsymbol{\theta}) + \mathbf{K}_k(\boldsymbol{\theta}) \frac{\partial \mathbf{v}_k(\boldsymbol{\theta})}{\partial \theta_i}, \\
 \frac{\partial \mathbf{P}_k(\boldsymbol{\theta})}{\partial \theta_i} &= \frac{\partial \mathbf{P}_k^-(\boldsymbol{\theta})}{\partial \theta_i} - \frac{\partial \mathbf{K}_k(\boldsymbol{\theta})}{\partial \theta_i} \mathbf{S}_k(\boldsymbol{\theta}) \mathbf{K}_k^\top(\boldsymbol{\theta}) \\
 &\quad - \mathbf{K}_k(\boldsymbol{\theta}) \frac{\partial \mathbf{S}_k(\boldsymbol{\theta})}{\partial \theta_i} \mathbf{K}_k^\top(\boldsymbol{\theta}) - \mathbf{K}_k(\boldsymbol{\theta}) \mathbf{S}_k(\boldsymbol{\theta}) \frac{\partial \mathbf{K}_k^\top(\boldsymbol{\theta})}{\partial \theta_i}.
 \end{aligned} \tag{A.68}$$

The recursion should be started from the initial condition $\partial \varphi_0(\boldsymbol{\theta})/\partial \boldsymbol{\theta} = -\partial \log p(\boldsymbol{\theta})/\partial \boldsymbol{\theta}$.

Another way to compute the same derivative is by using Fisher's identity (Equation 16.32) together with the expression for $\mathcal{Q}$ in Theorem 16.11. The result is the following.

Theorem A.11 (Energy function derivative for linear Gaussian model II). *The derivative of the energy function given in Theorem 16.9 can be computed as*

$$\frac{\partial \varphi_T(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = -\frac{\partial \log p(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} - \left. \frac{\partial \mathcal{Q}(\boldsymbol{\theta}, \boldsymbol{\theta}^{(n)})}{\partial \boldsymbol{\theta}} \right|_{\boldsymbol{\theta}^{(n)} = \boldsymbol{\theta}}, \tag{A.69}$$

where

$$\begin{aligned}
 & \left. \frac{\partial \mathcal{Q}(\boldsymbol{\theta}, \boldsymbol{\theta}^{(n)})}{\partial \theta_i} \right|_{\boldsymbol{\theta}^{(n)} = \boldsymbol{\theta}} \\
 &= -\frac{1}{2} \operatorname{tr} \left(\mathbf{P}_0^{-1} \frac{\partial \mathbf{P}_0}{\partial \theta_i} \right) - \frac{T}{2} \operatorname{tr} \left(\mathbf{Q}^{-1} \frac{\partial \mathbf{Q}}{\partial \theta_i} \right) - \frac{T}{2} \operatorname{tr} \left(\mathbf{R}^{-1} \frac{\partial \mathbf{R}}{\partial \theta_i} \right) \\
 &+ \frac{1}{2} \operatorname{tr} \left\{ \mathbf{P}_0^{-1} \frac{\partial \mathbf{P}_0}{\partial \theta_i} \mathbf{P}_0^{-1} \left[\mathbf{P}_0^s + (\mathbf{m}_0^s - \mathbf{m}_0) (\mathbf{m}_0^s - \mathbf{m}_0)^\top \right] \right\} \\
 &+ \frac{1}{2} \operatorname{tr} \left\{ \mathbf{P}_0^{-1} \left[\frac{\partial \mathbf{m}_0}{\partial \theta_i} (\mathbf{m}_0^s - \mathbf{m}_0)^\top + (\mathbf{m}_0^s - \mathbf{m}_0) \frac{\partial \mathbf{m}_0^\top}{\partial \theta_i} \right] \right\} \\
 &+ \frac{T}{2} \operatorname{tr} \left\{ \mathbf{Q}^{-1} \frac{\partial \mathbf{Q}}{\partial \theta_i} \mathbf{Q}^{-1} \left[\boldsymbol{\Sigma} - \mathbf{C} \mathbf{A}^\top - \mathbf{A} \mathbf{C}^\top + \mathbf{A} \boldsymbol{\Phi} \mathbf{A}^\top \right] \right\} \\
 &- \frac{T}{2} \operatorname{tr} \left\{ \mathbf{Q}^{-1} \left[-\mathbf{C} \frac{\partial \mathbf{A}^\top}{\partial \theta_i} - \frac{\partial \mathbf{A}}{\partial \theta_i} \mathbf{C}^\top + \frac{\partial \mathbf{A}}{\partial \theta_i} \boldsymbol{\Phi} \mathbf{A}^\top + \mathbf{A} \boldsymbol{\Phi} \frac{\partial \mathbf{A}^\top}{\partial \theta_i} \right] \right\} \\
 &+ \frac{T}{2} \operatorname{tr} \left\{ \mathbf{R}^{-1} \frac{\partial \mathbf{R}}{\partial \theta_i} \mathbf{R}^{-1} \left[\mathbf{D} - \mathbf{B} \mathbf{H}^\top - \mathbf{H} \mathbf{B}^\top + \mathbf{H} \boldsymbol{\Sigma} \mathbf{H}^\top \right] \right\} \\
 &- \frac{T}{2} \operatorname{tr} \left\{ \mathbf{R}^{-1} \left[-\mathbf{B} \frac{\partial \mathbf{H}^\top}{\partial \theta_i} - \frac{\partial \mathbf{H}}{\partial \theta_i} \mathbf{B}^\top + \frac{\partial \mathbf{H}}{\partial \theta_i} \boldsymbol{\Sigma} \mathbf{H}^\top + \mathbf{H} \boldsymbol{\Sigma} \frac{\partial \mathbf{H}^\top}{\partial \theta_i} \right] \right\}, \\
 & \tag{A.70}
 \end{aligned}$$

where all the terms are evaluated at $\boldsymbol{\theta}$.

A.11 Parameter Derivatives for the Gaussian Filter

In this section we consider the computation of the gradient of the Gaussian filtering-based energy function in Algorithm 16.14, which was considered with models of the form

$$\begin{aligned}
 \mathbf{x}_k &= \mathbf{f}(\mathbf{x}_{k-1}, \boldsymbol{\theta}) + \mathbf{q}_{k-1}, \\
 \mathbf{y}_k &= \mathbf{h}(\mathbf{x}_k, \boldsymbol{\theta}) + \mathbf{r}_k.
 \end{aligned}
 \tag{A.71}$$

In order to compute the derivative, it is convenient to first rewrite the expectations as expectations over unit Gaussian distributions as follows:

$$\begin{aligned}
 \mathbf{m}_k^-(\boldsymbol{\theta}) &= \int \mathbf{f}(\mathbf{x}_{k-1}, \boldsymbol{\theta}) \mathcal{N}(\mathbf{x}_{k-1} \mid \mathbf{m}_{k-1}(\boldsymbol{\theta}), \mathbf{P}_{k-1}(\boldsymbol{\theta})) \, d\mathbf{x}_{k-1} \\
 &= \int \mathbf{f}(\mathbf{m}_{k-1}(\boldsymbol{\theta}) + \sqrt{\mathbf{P}_{k-1}(\boldsymbol{\theta})} \boldsymbol{\xi}, \boldsymbol{\theta}) \mathcal{N}(\boldsymbol{\xi} \mid \mathbf{0}, \mathbf{I}) \, d\boldsymbol{\xi}.
 \end{aligned}
 \tag{A.72}$$

The derivative of this expression can now be computed as

$$\begin{aligned} \frac{\partial \mathbf{m}_k^-(\boldsymbol{\theta})}{\partial \theta_i} = \int & \left[\mathbf{F}_x(\mathbf{m}_{k-1}(\boldsymbol{\theta}) + \sqrt{\mathbf{P}_{k-1}(\boldsymbol{\theta})} \boldsymbol{\xi}, \boldsymbol{\theta}) \right. \\ & \times \left(\frac{\partial \mathbf{m}_{k-1}(\boldsymbol{\theta})}{\partial \theta_i} + \frac{\partial \sqrt{\mathbf{P}_{k-1}(\boldsymbol{\theta})}}{\partial \theta_i} \boldsymbol{\xi} \right) \\ & \left. + \frac{\partial \mathbf{f}}{\partial \theta_i}(\mathbf{m}_{k-1}(\boldsymbol{\theta}) + \sqrt{\mathbf{P}_{k-1}(\boldsymbol{\theta})} \boldsymbol{\xi}, \boldsymbol{\theta}) \right] \mathbf{N}(\boldsymbol{\xi} | \mathbf{0}, \mathbf{I}) \, d\boldsymbol{\xi}. \end{aligned}$$

Assuming that we are using the Cholesky factorization-based matrix square root, the derivative $\partial \sqrt{\mathbf{P}_{k-1}(\boldsymbol{\theta})} / \partial \theta_i$ can be computed with Algorithm A.7 or Theorem A.8 given in Section A.3.

The above form is directly suitable for sigma point methods, because they are based on the same change of variables. That is, the corresponding derivative of the sigma point approximation will be

$$\begin{aligned} \frac{\partial \mathbf{m}_k^-(\boldsymbol{\theta})}{\partial \theta_i} \approx \sum_j W_j & \left[\mathbf{F}_x(\mathbf{m}_{k-1}(\boldsymbol{\theta}) + \sqrt{\mathbf{P}_{k-1}(\boldsymbol{\theta})} \boldsymbol{\xi}^{(j)}, \boldsymbol{\theta}) \right. \\ & \times \left(\frac{\partial \mathbf{m}_{k-1}(\boldsymbol{\theta})}{\partial \theta_i} + \frac{\partial \sqrt{\mathbf{P}_{k-1}(\boldsymbol{\theta})}}{\partial \theta_i} \boldsymbol{\xi}^{(j)} \right) \\ & \left. + \frac{\partial \mathbf{f}}{\partial \theta_i}(\mathbf{m}_{k-1}(\boldsymbol{\theta}) + \sqrt{\mathbf{P}_{k-1}(\boldsymbol{\theta})} \boldsymbol{\xi}^{(j)}, \boldsymbol{\theta}) \right], \end{aligned} \quad (\text{A.73})$$

where W_j and $\boldsymbol{\xi}^{(j)}$ are the weights and unit sigma points of the used sigma point method. A nice thing in the above expression is that it is exactly the derivative of the sigma point approximation to $\mathbf{m}_k^-(\boldsymbol{\theta})$.

The derivatives of the Gaussian filtering-based energy function can now be expressed as follows.

Algorithm A.12 (Derivatives of Gaussian filtering-based energy function).
The recursion for the derivative of the approximate energy function is

$$\begin{aligned} \frac{\partial \varphi_k(\boldsymbol{\theta})}{\partial \theta_i} \simeq & \frac{\partial \varphi_{k-1}(\boldsymbol{\theta})}{\partial \theta_i} + \frac{1}{2} \operatorname{tr} \left(\mathbf{S}_k^{-1}(\boldsymbol{\theta}) \frac{\partial \mathbf{S}_k(\boldsymbol{\theta})}{\partial \theta_i} \right) + \mathbf{v}_k^\top(\boldsymbol{\theta}) \mathbf{S}_k^{-1}(\boldsymbol{\theta}) \frac{\partial \mathbf{v}_k(\boldsymbol{\theta})}{\partial \theta_i} \\ & - \frac{1}{2} \mathbf{v}_k^\top(\boldsymbol{\theta}) \mathbf{S}_k^{-1}(\boldsymbol{\theta}) \frac{\partial \mathbf{S}_k(\boldsymbol{\theta})}{\partial \theta_i} \mathbf{S}_k^{-1}(\boldsymbol{\theta}) \mathbf{v}_k(\boldsymbol{\theta}). \end{aligned} \quad (\text{A.74})$$

The derivatives of the prediction step are

$$\begin{aligned}
 \frac{\partial \mathbf{m}_k^-}{\partial \theta_i} &= \int \left[\mathbf{F}_x \left(\mathbf{m}_{k-1} + \sqrt{\mathbf{P}_{k-1}} \boldsymbol{\xi}, \boldsymbol{\theta} \right) \left(\frac{\partial \mathbf{m}_{k-1}}{\partial \theta_i} + \frac{\partial \sqrt{\mathbf{P}_{k-1}}}{\partial \theta_i} \boldsymbol{\xi} \right) \right. \\
 &\quad \left. + \frac{\partial \mathbf{f}}{\partial \theta_i} \left(\mathbf{m}_{k-1} + \sqrt{\mathbf{P}_{k-1}} \boldsymbol{\xi}, \boldsymbol{\theta} \right) \right] \mathbf{N}(\boldsymbol{\xi} | \mathbf{0}, \mathbf{I}) \, d\boldsymbol{\xi}, \\
 \frac{\partial \mathbf{P}_k^-}{\partial \theta_i} &= \int \left[\mathbf{F}_x \left(\mathbf{m}_{k-1} + \sqrt{\mathbf{P}_{k-1}} \boldsymbol{\xi}, \boldsymbol{\theta} \right) \left(\frac{\partial \mathbf{m}_{k-1}}{\partial \theta_i} + \frac{\partial \sqrt{\mathbf{P}_{k-1}}}{\partial \theta_i} \boldsymbol{\xi} \right) \right. \\
 &\quad \left. + \frac{\partial \mathbf{f}}{\partial \theta_i} \left(\mathbf{m}_{k-1} + \sqrt{\mathbf{P}_{k-1}} \boldsymbol{\xi}, \boldsymbol{\theta} \right) - \frac{\partial \mathbf{m}_k^-}{\partial \theta_i} \right] \\
 &\quad \times \left[\mathbf{f} \left(\mathbf{m}_{k-1} + \sqrt{\mathbf{P}_{k-1}} \boldsymbol{\xi}, \boldsymbol{\theta} \right) - \mathbf{m}_k^- \right]^\top \mathbf{N}(\boldsymbol{\xi} | \mathbf{0}, \mathbf{I}) \, d\boldsymbol{\xi} \\
 &\quad + \int \left[\mathbf{f} \left(\mathbf{m}_{k-1} + \sqrt{\mathbf{P}_{k-1}} \boldsymbol{\xi}, \boldsymbol{\theta} \right) - \mathbf{m}_k^- \right] \\
 &\quad \times \left[\mathbf{F}_x \left(\mathbf{m}_{k-1} + \sqrt{\mathbf{P}_{k-1}} \boldsymbol{\xi}, \boldsymbol{\theta} \right) \left(\frac{\partial \mathbf{m}_{k-1}}{\partial \theta_i} + \frac{\partial \sqrt{\mathbf{P}_{k-1}}}{\partial \theta_i} \boldsymbol{\xi} \right) \right. \\
 &\quad \left. + \frac{\partial \mathbf{f}}{\partial \theta_i} \left(\mathbf{m}_{k-1} + \sqrt{\mathbf{P}_{k-1}} \boldsymbol{\xi}, \boldsymbol{\theta} \right) - \frac{\partial \mathbf{m}_k^-}{\partial \theta_i} \right]^\top \mathbf{N}(\boldsymbol{\xi} | \mathbf{0}, \mathbf{I}) \, d\boldsymbol{\xi} + \frac{\partial \mathbf{Q}_{k-1}}{\partial \theta_i},
 \end{aligned} \tag{A.75}$$

where $\mathbf{F}_x$ is the Jacobian matrix of $\mathbf{x} \mapsto \mathbf{f}(\mathbf{x}, \boldsymbol{\theta})$. In the above expressions as well as in the following, we have dropped the dependencies of various terms on the parameters $\boldsymbol{\theta}$ to simplify the notation. The derivatives of the update step are

$$\begin{aligned}
 \frac{\partial \boldsymbol{\mu}_k}{\partial \theta_i} &= \int \left[\mathbf{H}_x \left(\mathbf{m}_k^- + \sqrt{\mathbf{P}_k^-} \boldsymbol{\xi}, \boldsymbol{\theta} \right) \left(\frac{\partial \mathbf{m}_k^-}{\partial \theta_i} + \frac{\partial \sqrt{\mathbf{P}_k^-}}{\partial \theta_i} \boldsymbol{\xi} \right) \right. \\
 &\quad \left. + \frac{\partial \mathbf{h}}{\partial \theta_i} \left(\mathbf{m}_k^- + \sqrt{\mathbf{P}_k^-} \boldsymbol{\xi}, \boldsymbol{\theta} \right) \right] \mathbf{N}(\boldsymbol{\xi} | \mathbf{0}, \mathbf{I}) \, d\boldsymbol{\xi}, \\
 \frac{\partial \mathbf{v}_k}{\partial \theta_i} &= -\frac{\partial \boldsymbol{\mu}_k}{\partial \theta_i}, \\
 \frac{\partial \mathbf{S}_k}{\partial \theta_i} &= \int \left[\mathbf{H}_x \left(\mathbf{m}_k^- + \sqrt{\mathbf{P}_k^-} \boldsymbol{\xi}, \boldsymbol{\theta} \right) \left(\frac{\partial \mathbf{m}_k^-}{\partial \theta_i} + \frac{\partial \sqrt{\mathbf{P}_k^-}}{\partial \theta_i} \boldsymbol{\xi} \right) \right.
 \end{aligned}$$

$$\begin{aligned}
& + \frac{\partial \mathbf{h}}{\partial \theta_i} \left(\mathbf{m}_k^- + \sqrt{\mathbf{P}_k^-} \boldsymbol{\xi}, \boldsymbol{\theta} \right) - \frac{\partial \boldsymbol{\mu}_k}{\partial \theta_i} \Bigg] \\
& \times \left[\mathbf{h} \left(\mathbf{m}_k^- + \sqrt{\mathbf{P}_k^-} \boldsymbol{\xi}, \boldsymbol{\theta} \right) - \boldsymbol{\mu}_k \right]^\top \mathbf{N}(\boldsymbol{\xi} \mid \mathbf{0}, \mathbf{I}) \, d\boldsymbol{\xi} \\
& + \int \left[\mathbf{h} \left(\mathbf{m}_k^- + \sqrt{\mathbf{P}_k^-} \boldsymbol{\xi}, \boldsymbol{\theta} \right) - \boldsymbol{\mu}_k \right] \\
& \times \left[\mathbf{H}_x \left(\mathbf{m}_k^- + \sqrt{\mathbf{P}_k^-} \boldsymbol{\xi}, \boldsymbol{\theta} \right) \left(\frac{\partial \mathbf{m}_k^-}{\partial \theta_i} + \frac{\partial \sqrt{\mathbf{P}_k^-}}{\partial \theta_i} \boldsymbol{\xi} \right) \right. \\
& \quad \left. + \frac{\partial \mathbf{h}}{\partial \theta_i} \left(\mathbf{m}_k^- + \sqrt{\mathbf{P}_k^-} \boldsymbol{\xi}, \boldsymbol{\theta} \right) - \frac{\partial \boldsymbol{\mu}_k}{\partial \theta_i} \right]^\top \mathbf{N}(\boldsymbol{\xi} \mid \mathbf{0}, \mathbf{I}) \, d\boldsymbol{\xi} + \frac{\partial \mathbf{R}_k}{\partial \theta_i}, \\
\frac{\partial \mathbf{C}_k}{\partial \theta_i} &= \int \left\{ \frac{\partial \sqrt{\mathbf{P}_k^-}}{\partial \theta_i} \boldsymbol{\xi} \left(\mathbf{h} \left(\mathbf{m}_k^- + \sqrt{\mathbf{P}_k^-} \boldsymbol{\xi}, \boldsymbol{\theta} \right) - \boldsymbol{\mu}_k \right)^\top \right. \\
& \quad + \sqrt{\mathbf{P}_k^-} \boldsymbol{\xi} \left[\mathbf{H}_x \left(\mathbf{m}_k^- + \sqrt{\mathbf{P}_k^-} \boldsymbol{\xi}, \boldsymbol{\theta} \right) \left(\frac{\partial \mathbf{m}_k^-}{\partial \theta_i} + \frac{\partial \sqrt{\mathbf{P}_k^-}}{\partial \theta_i} \boldsymbol{\xi} \right) \right. \\
& \quad \left. \left. + \frac{\partial \mathbf{h}}{\partial \theta_i} \left(\mathbf{m}_k^- + \sqrt{\mathbf{P}_k^-} \boldsymbol{\xi}, \boldsymbol{\theta} \right) - \frac{\partial \boldsymbol{\mu}_k}{\partial \theta_i} \right]^\top \right\} \mathbf{N}(\boldsymbol{\xi} \mid \mathbf{0}, \mathbf{I}) \, d\boldsymbol{\xi}, \\
\frac{\partial \mathbf{K}_k}{\partial \theta_i} &= \frac{\partial \mathbf{C}_k}{\partial \theta_i} \mathbf{S}_k^{-1} - \mathbf{C}_k \mathbf{S}_k^{-1} \frac{\partial \mathbf{S}_k}{\partial \theta_i} \mathbf{S}_k^{-1}, \\
\frac{\partial \mathbf{m}_k}{\partial \theta_i} &= \frac{\partial \mathbf{m}_k^-}{\partial \theta_i} + \frac{\partial \mathbf{K}_k}{\partial \theta_i} \mathbf{v}_k + \mathbf{K}_k \frac{\partial \mathbf{v}_k}{\partial \theta_i}, \\
\frac{\partial \mathbf{P}_k}{\partial \theta_i} &= \frac{\partial \mathbf{P}_k^-}{\partial \theta_i} - \frac{\partial \mathbf{K}_k}{\partial \theta_i} \mathbf{S}_k \mathbf{K}_k^\top - \mathbf{K}_k \frac{\partial \mathbf{S}_k}{\partial \theta_i} \mathbf{K}_k^\top - \mathbf{K}_k \mathbf{S}_k \frac{\partial \mathbf{K}_k^\top}{\partial \theta_i},
\end{aligned} \tag{A.76}$$

where $\mathbf{H}_x$ is the Jacobian matrix of $\mathbf{x} \mapsto \mathbf{h}(\mathbf{x}, \boldsymbol{\theta})$. The derivatives of the Cholesky factors can be computed with Algorithm A.7 or Theorem A.8 given in Section A.3.

The corresponding sigma point approximations can be obtained by approximating the integrals analogously to Equation (A.73). The resulting derivative will be exact in the sense that it is the exact derivative of the corresponding sigma point-based approximation to the energy function.

We could also change back to the original variable, which gives, for example, the following representation for the derivative of the predicted

mean:

$$\frac{\partial \mathbf{m}_k^-}{\partial \theta_i} = \int \left[\mathbf{F}_x(\mathbf{x}_{k-1}, \boldsymbol{\theta}) \mathbf{g}(\mathbf{x}_{k-1}, \boldsymbol{\theta}) + \frac{\partial \mathbf{f}(\mathbf{x}_{k-1}, \boldsymbol{\theta})}{\partial \theta_i} \right] \times N(\mathbf{x}_{k-1} \mid \mathbf{m}_{k-1}, \mathbf{P}_{k-1}) \, d\mathbf{x}_{k-1}, \quad (\text{A.77})$$

where

$$\begin{aligned} & \mathbf{g}(\mathbf{x}_{k-1}, \boldsymbol{\theta}) \\ &= \frac{\partial \mathbf{m}_{k-1}}{\partial \theta_i} + \frac{\partial \sqrt{\mathbf{P}_{k-1}}}{\partial \theta_i} \left(\sqrt{\mathbf{P}_{k-1}} \right)^{-1} (\mathbf{x}_{k-1} - \mathbf{m}_{k-1}). \end{aligned} \quad (\text{A.78})$$

The derivation of the full set of equations is left as an exercise for the reader.